67978: A Needle in a Data Haystack

What Intake Records Actually Tell You

Prediction, Adoption Delays, and Stray Reunification



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Domain

Municipal shelters receive stray, surrendered, and transferred animals, then record outcomes such as adoption, return to owner, transfer, or euthanasia. We ask whether intake records can (1) predict outcome type, (2) reveal traits associated with adoption delay, and (3) explain spatial patterns in stray-dog reunification. In all three problems, the raw pattern changes once we adjust for other variables or test whether it holds on new data.

Data

Our main data are two linked Austin Animal Center tables from the City of Austin Open Data Portal. We use Sonoma County Animal Services data as an independent dataset when its fields are comparable.

Dataset	Raw data	Structure and fields used	Coverage
Austin Outcomes	173,775 rows 25.2 MB	Outcome record per row. 12 total columns. Animal ID, outcome type, outcome time, species, age, sex, breed, and color.	Oct. 2013–May 2025
Austin Intakes	173,812 rows 26.0 MB	Intake record per row. 12 total columns. Animal ID, intake type, intake time, condition, found location, species, age, sex, breed, and color.	Oct. 2013–May 2025
Sonoma stays	33,718 rows 7.3 MB	Shelter stay per row. 24 total columns. Animal ID, animal type, breed, color, sex, birth date, intake date, outcome date, intake type, outcome type, condition, and days in shelter.	Intake: Aug. 2013–July 2026

Preprocessing. The outcomes table lacks intake dates and conditions, so we match each outcome to the same animal's most recent *preceding* intake, matching 172,246/173,775 outcomes (99.1%). After excluding 79 stays above 730 days, 172,167 rows remain (mean 20.7, median 6.0 days), and Problem 2 checks this cutoff. Of 156,287 unique IDs, 13,487 appear in more than one stay. Only 214 rows (0.12%) come from animal–intake-time combinations linked to two outcomes, so we keep them.

For Problem 3, eligible `Found Location` values are additionally geocoded using the U.S. Census batch geocoder. The filtering and match rate are described with that analysis.

Sonoma preprocessing. To make its fields comparable with Austin, we align outcome, intake, and condition labels using fixed mappings. We then require nonmissing intake and outcome dates, 0–730 shelter days, and nonnegative age derived from birth date. Birth date is missing in 23.7% of raw rows.

Final analytic cohorts. Because the three problems apply different task-specific filters, there is no single final Austin cohort. Problem 1 uses 172,157 stays, Problem 2 uses 83,896 adopted stays, and Problem 3 uses 26,045 geocoded stray dogs in the 39 ZIPs retained for analysis. The filtered Sonoma set contains 25,508 stays.

Data limitations. These administrative records describe animals entering the participating shelters rather than the general animal population. Early Austin outcomes may lack a preceding intake, and differences in shelter recording practices limit direct cross-city comparison.

Sample flow. Figure 1 summarizes the preprocessing steps and the resulting analysis cohorts.

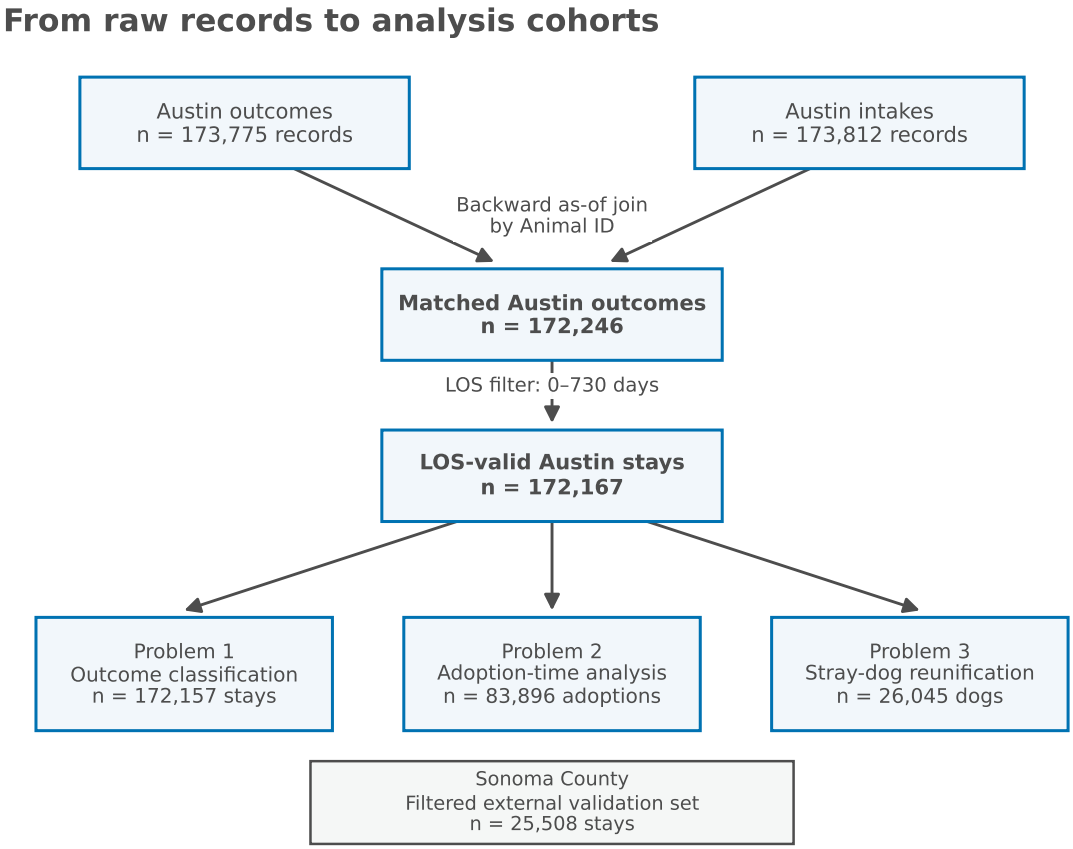


Figure 1: The three Austin analyses use different subsets of the matched and filtered data. Sonoma is kept separate as a filtered external validation set.

Problem 1: Outcome-Type Classification

Problem Description

Using only information available at intake, can we predict whether an animal will be adopted, returned to its owner, transferred, or have another outcome?

Our Solution

We retain Adoption, Return to Owner, and Transfer as separate classes. All remaining or missing outcome types are pooled into Other.

Our intake-time features include species, sex, altered (spayed/neutered) status, intake type, condition group, black-coat flag, month, age, occupancy, and trailing seven-day intake volume. Occupancy counts animals already present at intake. Seven-day volume counts intakes in $[t - 7 \text{ days}, t)$, so both use only pre-intake information. Fourteen condition labels are pooled into Normal, Injured, Medical, and Behavior/Other.

We treat this as a supervised four-class classification problem. Balanced L_2-regularized multinomial logistic regression is our interpretable linear baseline, and histogram-based gradient boosting is our nonlinear model. Boosting can capture feature interactions that the linear model cannot. Categorical features are one-hot encoded. Numeric features are unchanged for gradient boosting and standardized for logistic regression.

For gradient boosting, balanced class weights address the class imbalance. We keep scikit-learn 1.9's default complexity controls `learning_rate=0.1`, `max_iter=100`, `max_leaf_nodes=31`,

`min_samples_leaf=20`, and `early_stopping=auto` to avoid extra tuning, and use random seed 42 for reproducibility.

Interactive demo. The Streamlit app (or demo video) shows outcome probabilities for user-specified intake profiles and is not intended as a shelter decision tool.

Evaluation

Evaluation Criteria

The classes are imbalanced: Adoption makes up 48.5% of the test set, while Other makes up 8.3%, so accuracy can hide weak performance on less common classes. We therefore use macro-F_1 as our main metric, report per-class precision and recall, inspect a row-normalized confusion matrix, and check generalization over time and on Sonoma.

Setup

An animal-level 80/20 split gives 137,734 training and 34,423 test rows, with no animal ID appearing in both sets. Feature groups are developed by five-fold animal-grouped cross-validation within the training data. The held-out test set is used only after this development.

Results & Visualization

Figure 2 shows that outcome distributions differ substantially by intake type and condition, supporting these intake-time variables as model features.

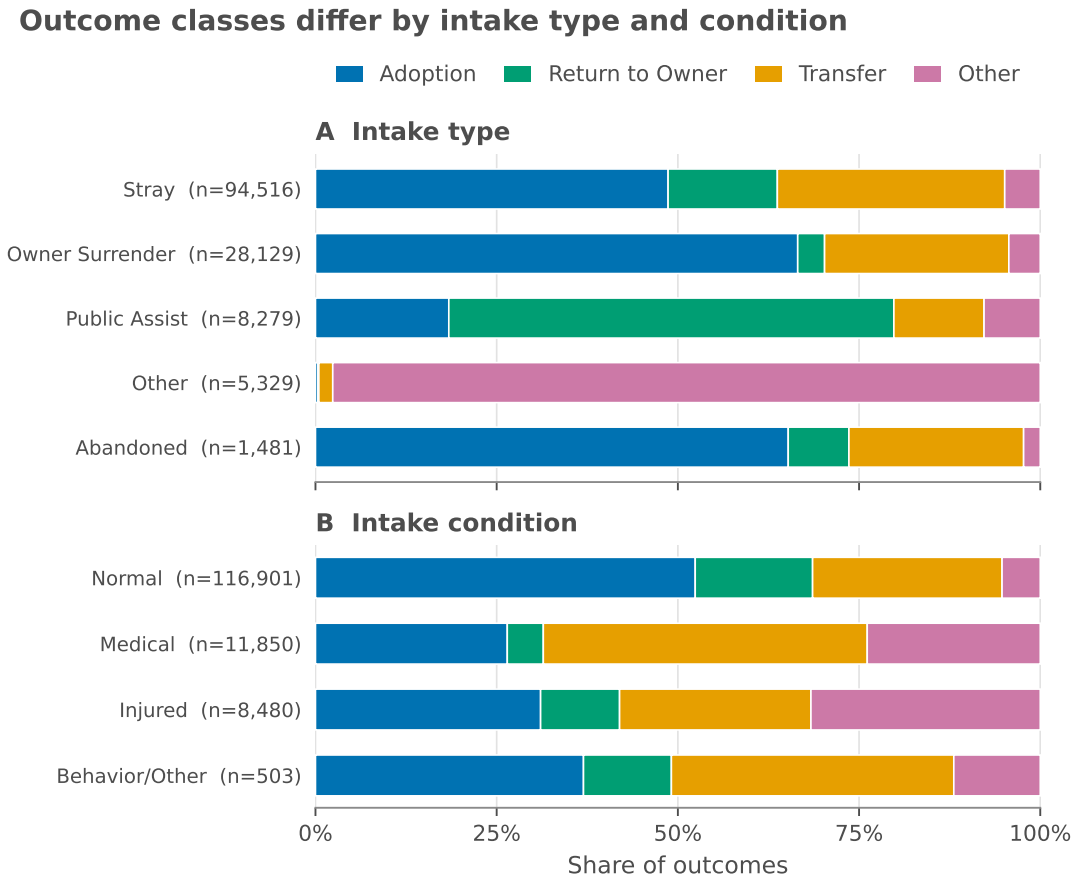


Figure 2: Training-set composition by intake type and condition. Each bar sums to 100%, with group sizes shown in labels.

Figure 3 shows that intake context gives the largest feature-group gain, while calendar and shelter-state features add smaller improvements. We then compare logistic regression and gradient boosting once on the untouched test set.

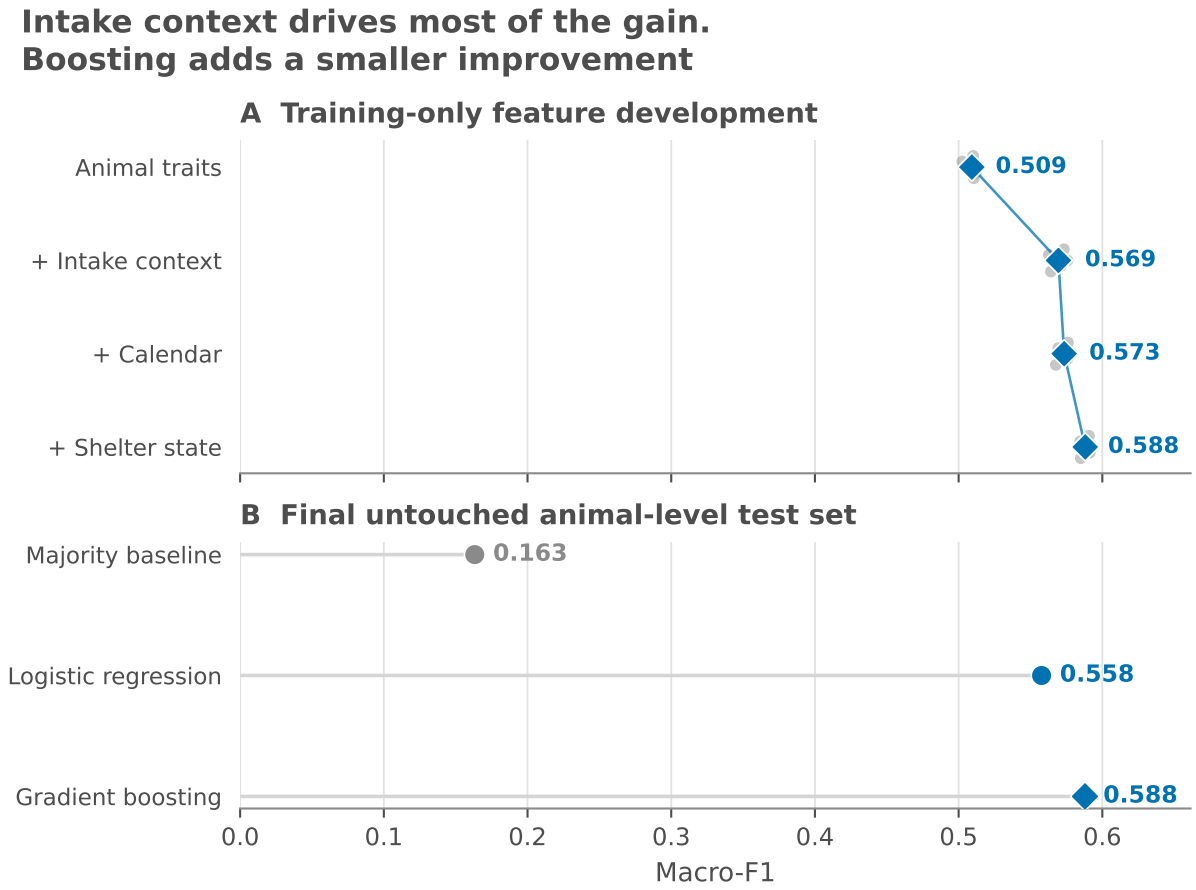


Figure 3: Top: five-fold animal-grouped feature development. Bottom: final untouched-test performance. Diamonds show fold means, small points show validation folds, and lines show mean ± one standard deviation.

Finding. On the untouched animal-level test set, boosting reaches macro-F_1 **0.588**, versus **0.558** for logistic regression and **0.163** for the majority baseline.

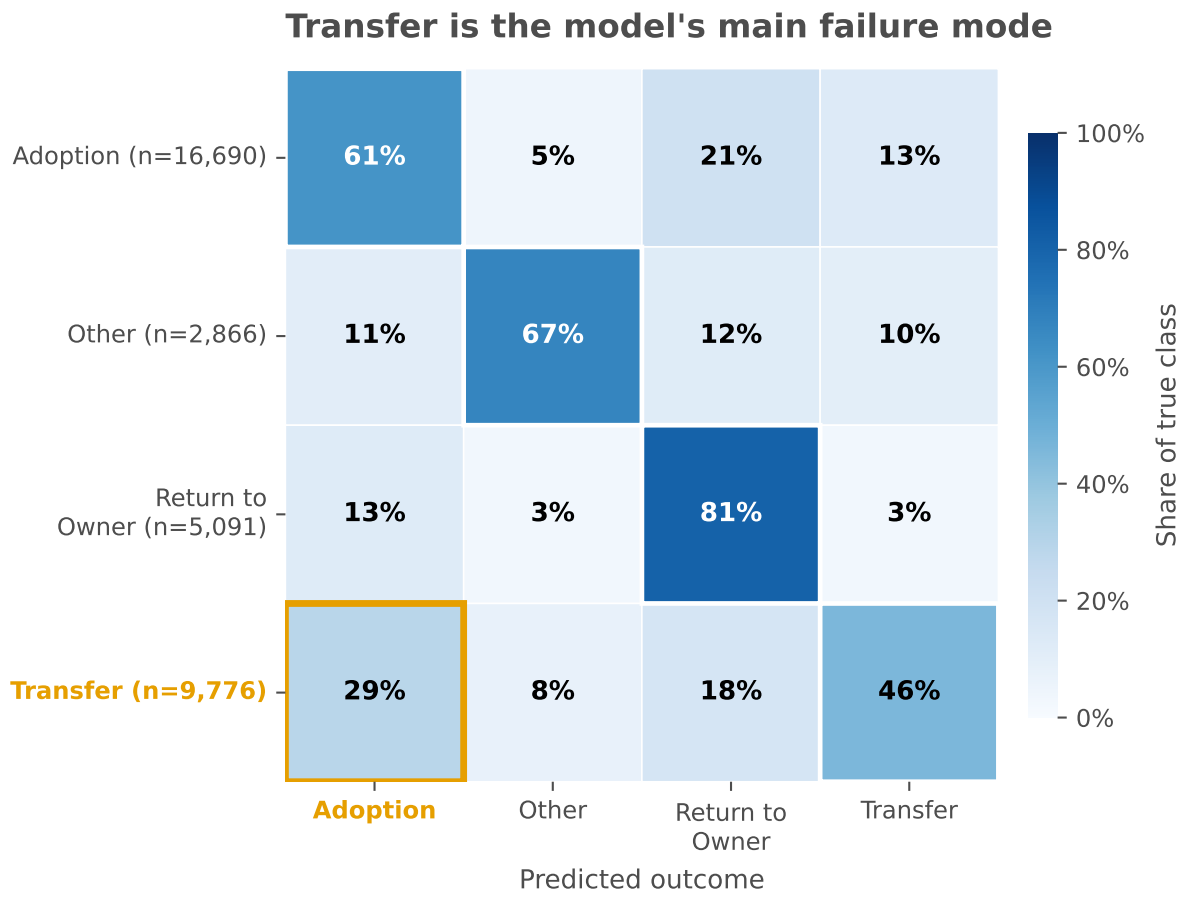


Figure 4: Rows are actual outcomes and columns are predicted outcomes. Values are percentages within each true class, so each row sums to 100%. The highlighted cell marks Transfer cases predicted as Adoption.

Return-to-Owner has high recall (0.81) but low precision (0.43), while Transfer has the lowest recall (0.46). Among Transfer cases, 29.0% are predicted as Adoption and 17.5% as Return to Owner.

Several of the highest-confidence errors involve animals other than cats and dogs being predicted as Other when their true outcome is Transfer. This suggests that some rescue-transfer decisions depend on information not recorded at intake. Macro-F_1 is 0.541 for cats, 0.432 for dogs, and 0.564 for other animals.

Robustness. Removing occupancy and recent-intake volume lowers Transfer recall from 0.46 to 0.42 and Macro-F_1 from 0.588 to 0.571.

Forward-time generalization. We train only on records before January 1, 2023 and evaluate on later animals, with no ID overlap. This gives roughly nine years of training data and more than two years of later outcomes. Macro-F_1 remains 0.557 versus a 0.189 training-majority baseline.

Cross-city generalization to Sonoma. The Austin-trained intake-only model generalizes better to Sonoma than the full feature model and remains above the Sonoma majority baseline (Figure 5). This suggests that calendar and shelter-state features are less portable across cities.

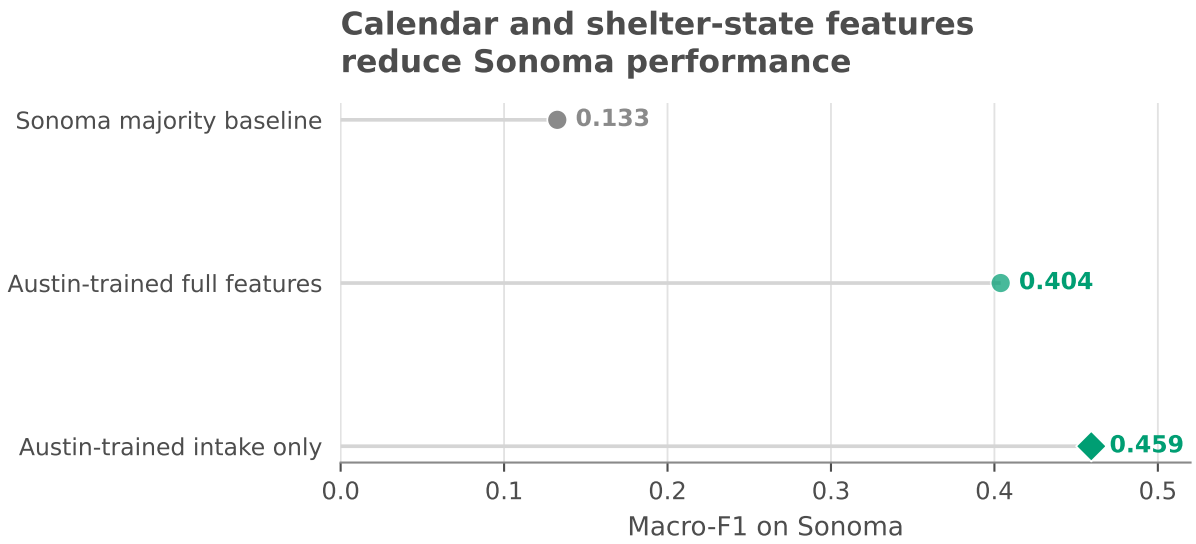


Figure 5: Macro-F_1 on the external Sonoma dataset for classifiers trained on Austin data.

Impediments

1. **Outcome classes are uneven.** The Other class combines several rare outcomes, while Transfer and Return-to-Owner are harder for the model to predict. To avoid hiding this behind overall accuracy, we report macro-F_1, per-class precision and recall, and the confusion matrix. We also avoid treating Other as equivalent to euthanasia.
2. **Intake data are incomplete.** Occupancy near the start of the dataset is incomplete because earlier animals are not observed, and later treatment or rescue-partner decisions are not recorded at intake. We therefore use only information available by intake, test forward-time and cross-city generalization, and present the model as a retrospective analysis rather than a shelter decision tool.

Problem 2: Traits Associated with Longer Waits Among Adoptees

Problem Description

Among animals eventually adopted, are shelter-recorded Pit Bull-type labels and black coloring associated with longer waits after accounting for observable intake differences?

Our Solution

We analyze matched Adoption outcomes with stays up to two years (730 days), and repeat the model with 365- and 1,095-day limits as sensitivity checks. Pit Bull-type is flagged when the recorded breed contains `pit bull`, and black coloring when the recorded color contains `black`, both ignoring case. Missing values are treated as nonmatches.

We model log(1 + adoption days) rather than raw days, which compresses very long waits and keeps zero-day adoptions defined. We fit OLS using both traits of interest plus age, species, sex, altered status, intake type, and condition. We report $100 \times (e^{\hat{\beta}} - 1)$, the percent difference in the model-implied geometric mean of 1 + adoption days. It is not a percentage change in expected raw adoption days.

Evaluation

Evaluation Criteria

With this sample size, even small effects can have small p-values, so we focus on effect sizes and 95% confidence intervals. We also check dog-only and cat-only models, different stay cutoffs, intake-year fixed effects, a quadratic age term, and an independent Sonoma replication.

Setup

Austin contributes 83,896 adoptions: 7,379 Pit Bull-type (8.8%) and 26,816 with black coloring (32.0%). Because some animals appear in more than one stay, we cluster regression standard errors by animal ID so repeated observations from the same animal are not treated as fully independent. The data contain 76,237 distinct animal IDs. Dog-only and cat-only models use 47,127 and 35,429 rows, respectively.

Results & Visualization

Figure 6 shows a large raw wait-time difference for Pit Bull-type adoptees, while black-color medians are nearly identical.

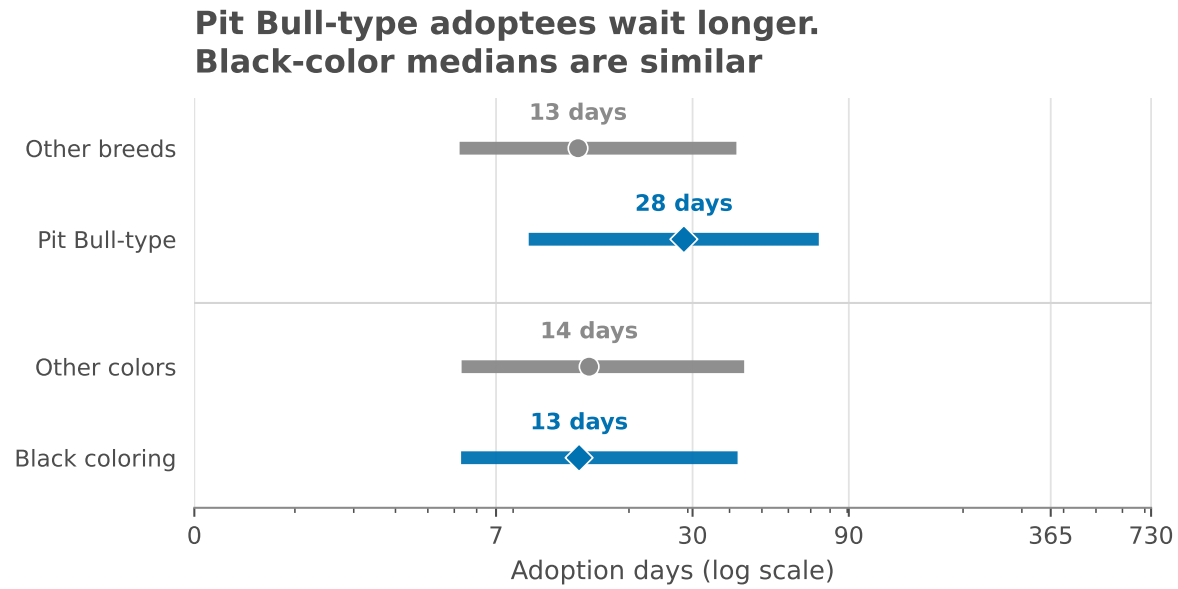


Figure 6: Points show median adoption times, and horizontal bars show the middle 50% of adoption times.

After adjustment, the Pit Bull-type association increases from +74.0% to +126.3%, while the black-color association changes from −2.9% to +10.2% (Figure 7). The adjusted 95% confidence intervals are +119.1% to +133.7% and +8.4% to +12.0%, respectively. The reversal for black coloring suggests that the raw comparison was masking differences associated with the other intake characteristics in the model.

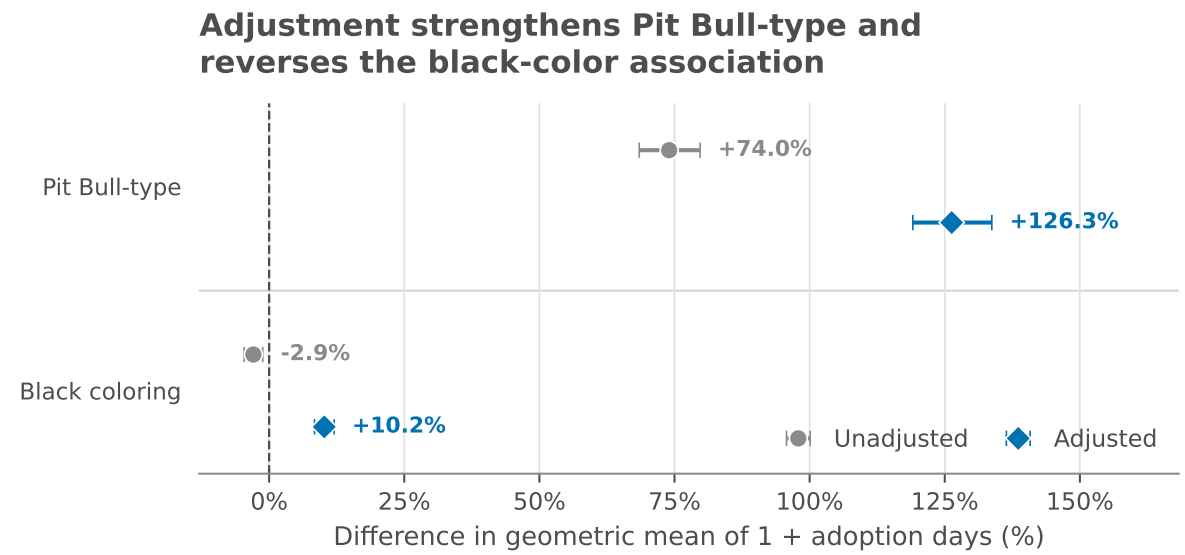


Figure 7: Points show unadjusted and adjusted adoption-time associations. Lines show animal-clustered 95% confidence intervals.

Robustness. Model $R^2 = 0.150$ on the log-transformed outcome, so most variation in adoption time remains unexplained by the measured variables. Within dogs, the adjusted Pit Bull-type association is +123.5%. Black coloring is +7.2% for dogs and +17.0% for cats. Changing the stay cutoff to 365 or 1,095 days gives Pit Bull-type estimates of +117.3% and +127.5%, while black coloring remains +10.2–10.3%. Adding intake-year fixed effects and a quadratic age term leaves the Austin estimates stable at +121.4% for Pit Bull-type and +10.5% for black coloring.

External replication. We refit the same OLS specification on 9,201 Sonoma adoptions. Both associations remain positive in Sonoma (Figure 8). The black-color estimate is similar across cities, while the Pit Bull-type association is much smaller in Sonoma.

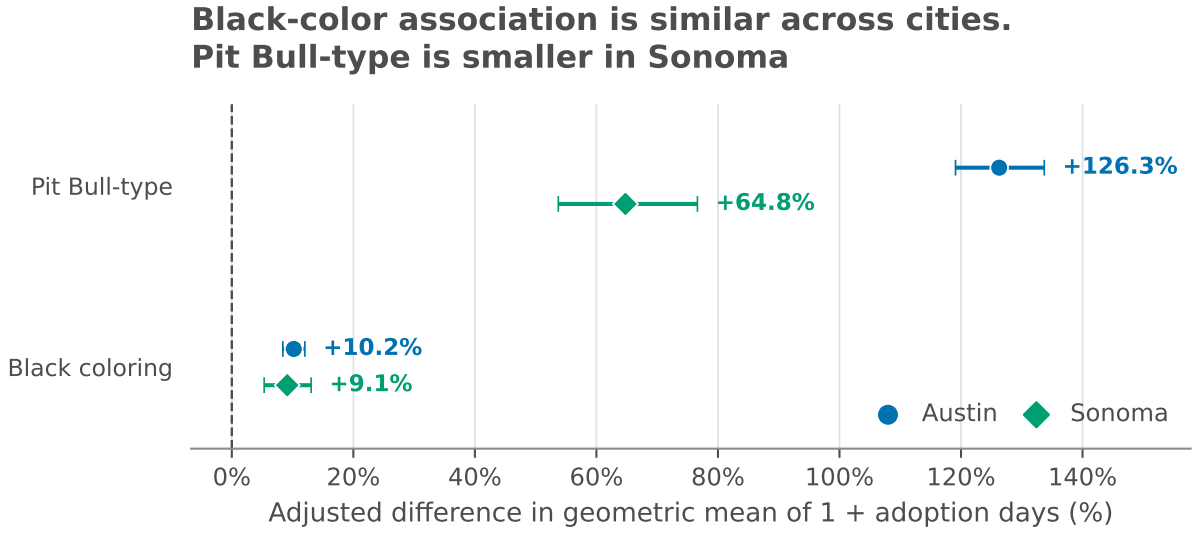


Figure 8: Adjusted adoption-time associations estimated separately in Austin and Sonoma. Lines show animal-clustered 95% confidence intervals.

Impediments

1. **Shelter labels are imperfect.** Pit Bull-type is based on the recorded shelter label, not a verified breed or genetic test.

2. **We study adoptees only.** The model asks how long adopted animals waited, not whether either trait changed the probability of adoption.
3. **Listing text was not reliable enough.** After obtaining RescueGroups API access, we found that only 9.7% of 952 adopted records had both usable text and a valid adoption duration. The API provides current rather than versioned descriptions, so we could not verify that the text was available before adoption and excluded it from the analysis.

Problem 3: Geography and Stray Reunification

Problem Description

Does where a stray dog is found predict its chance of being reclaimed, and how much of the apparent geographic variation remains after accounting for measured dog characteristics?

Our Solution

`Found Location` contains a specific location for 95.8% of stray intakes but only 13.9% of owner surrenders. Owner-surrender records often contain a jurisdiction placeholder such as `Austin (TX)` instead, so we restrict the geographic analysis to strays. For Census geocoding, we further keep numbered Austin street addresses. The U.S. Census batch geocoder matches 29,461 of 33,351 unique numbered street addresses (88.3%) to ZIP, tract, and coordinates.

We focus on dogs because reunification differs sharply by species, at 26.4% for dogs versus 3.6% for cats. We compare a logistic regression using age, altered status at intake, Pit Bull-type label, and condition with the same model plus ZIP. Altered status is recorded at intake, so it is not outcome leakage.

To check how much geographic structure remains after adjustment, we compare observed reunification with the rate expected by the traits model on a hexagonal grid. We also use DBSCAN on pickup coordinates as a check that does not depend on ZIP or grid boundaries.

Evaluation

Evaluation Criteria

We test whether adding ZIP improves prediction on held-out dogs. Log loss penalizes confident wrong probabilities, Brier score measures squared probability error, and AUC measures ranking quality. Lower is better for log loss and Brier score, while higher is better for AUC. We compare the traits-only and traits-plus-ZIP models and show 95% confidence intervals for ZIP-level reunification rates.

Setup

There are 60,861 geocoded strays across 52 ZIPs and 278 tracts. The main models use 26,045 dogs in 39 ZIPs. We require at least 60 dogs per reported ZIP to avoid very unstable rates. At this sample size and near the overall reunification rate, the approximate 95% margin is about ±11 percentage points. Regression standard errors are clustered by animal ID, with 24,886 distinct dogs. Five 80/20 grouped holdouts keep each animal entirely in either train or test.

Results & Visualization

Unadjusted reunification varies about threefold across the 39 Austin ZIPs with at least 60 dogs, showing a strong raw geographic pattern before adjustment (Figure 9).

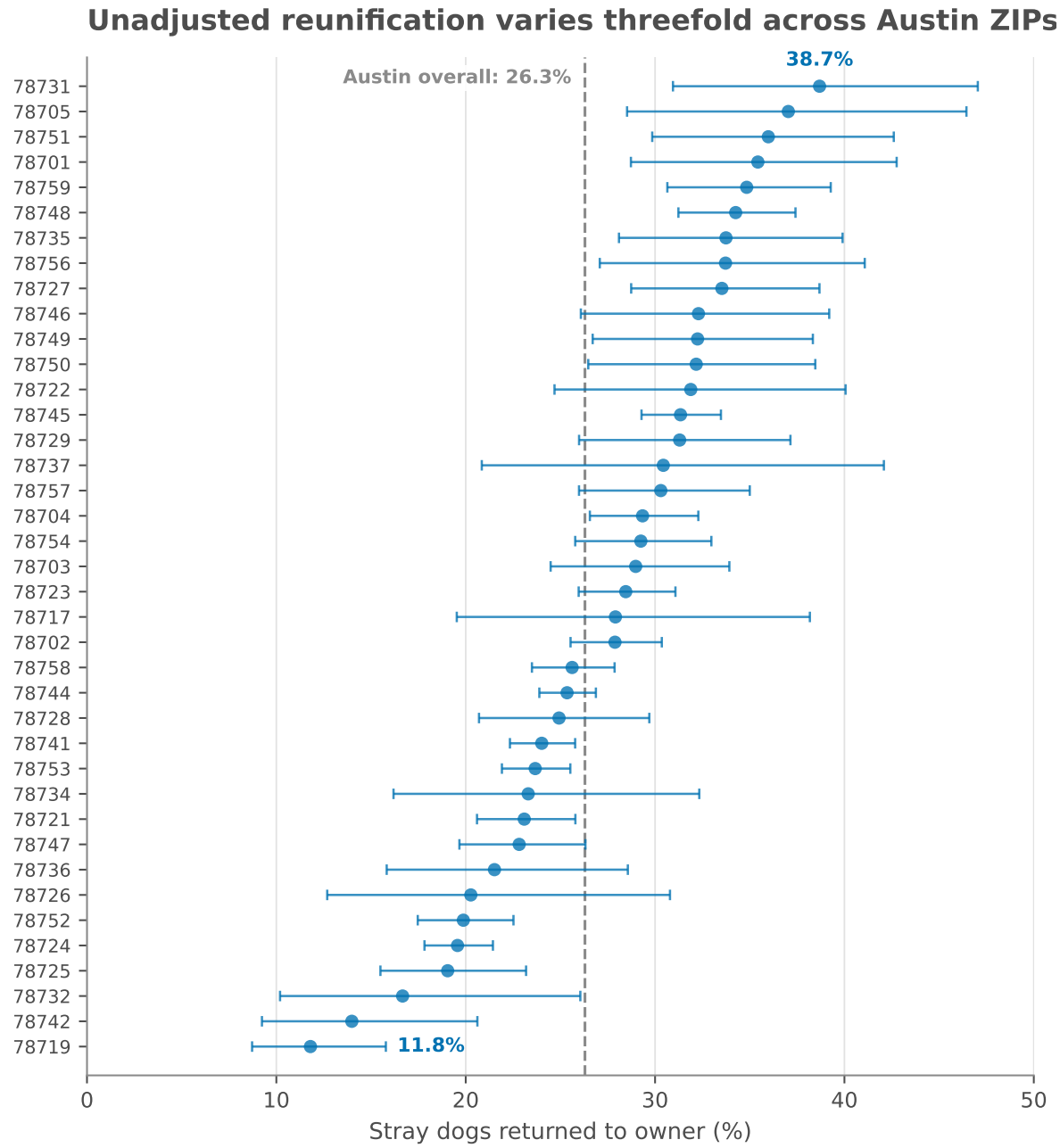


Figure 9: Points show ZIP-level stray-dog reunification rates with 95% confidence intervals. The dashed line shows the Austin-wide rate.

Finding. Altered dogs return at **54.0%**, versus **16.8%** for intact dogs, which helps account for much of the geographic pattern.

Species mix shows little relationship with reunification across the 40 ZIPs with at least 150 geocoded strays (% dogs, $r = -0.066$). In contrast, altered share among dogs is strongly related across the 39 ZIPs with at least 60 dogs ($r = 0.784$, dog-count-weighted $r = 0.818$, Figure 10).

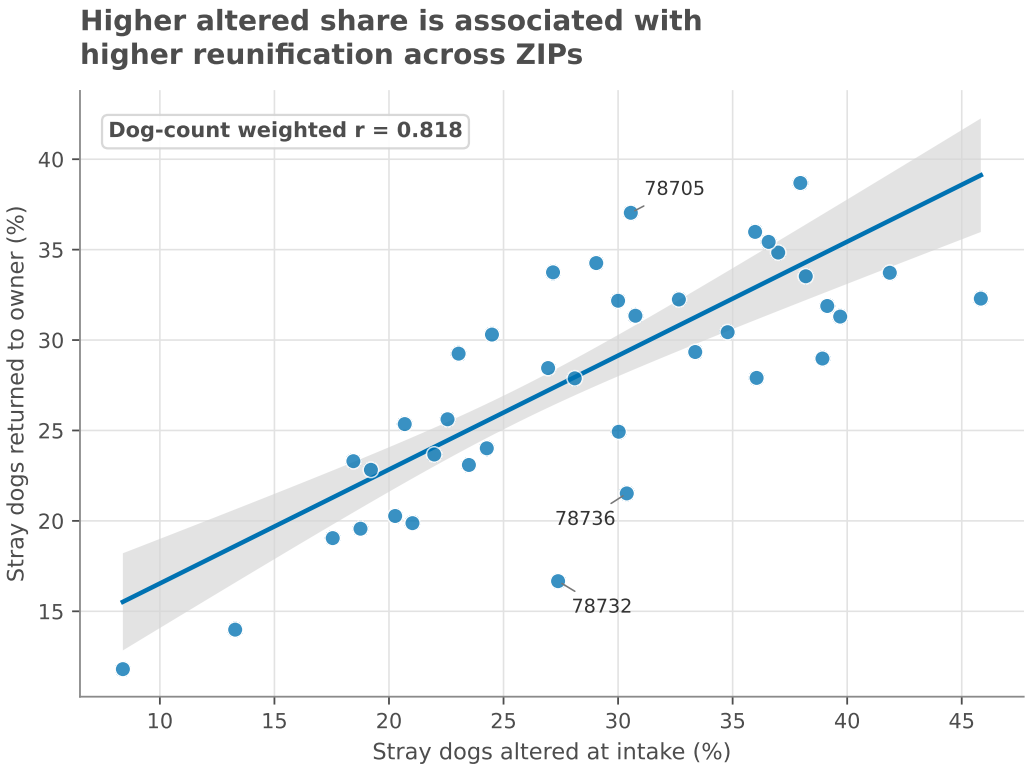


Figure 10: Each point is one of the 39 ZIPs with at least 60 dogs. The line is a dog-count-weighted fit with a 95% confidence band. The association is descriptive.

Altered status remains strongly associated with reunification after adjustment. Intact dogs have about one-quarter the odds of reunification compared with altered dogs (odds ratio 0.255, 95% CI 0.238–0.273). Equivalently, altered dogs have about 3.9 times the odds of reunification.

Held-out prediction. Adding ZIP improves in-sample fit, but the grouped held-out gain is very small. Across five grouped holdouts, log loss changes from 0.4934 to 0.4921 and Brier score from 0.1608 to 0.1604, while AUC slightly decreases from 0.7584 to 0.7557. Overall, location adds little predictive value once the measured dog traits are known.

At a finer spatial scale, we divide Austin into a hexagonal grid and report cells with at least 60 dogs. We use a 14-column grid for the main figures, and the adjustment pattern remains similar for grids from 10 to 18 columns.

After adjustment, the standard deviation of cell-level geographic differences, weighted by the number of dogs in each cell, falls from 6.2 to 4.3 percentage points. In 57 of 80 cells with at least 60 dogs (71%), the deviation moves closer to zero, covering 72.7% of the displayed dogs. Figure 11 summarizes this reduction, while Figure 12 shows where the remaining differences occur.

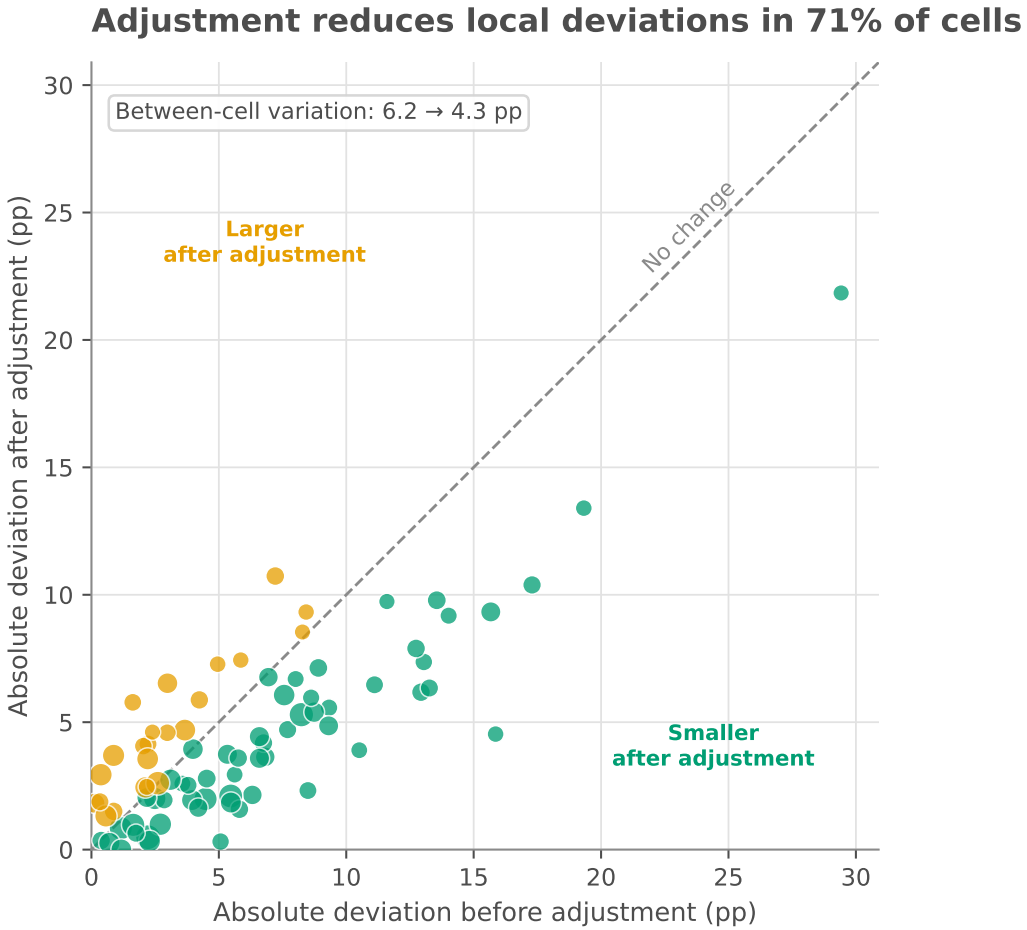


Figure 11: Absolute observed versus trait-adjusted geographic deviations. Points below the identity line move closer to zero after adjustment, and point area represents dog count.

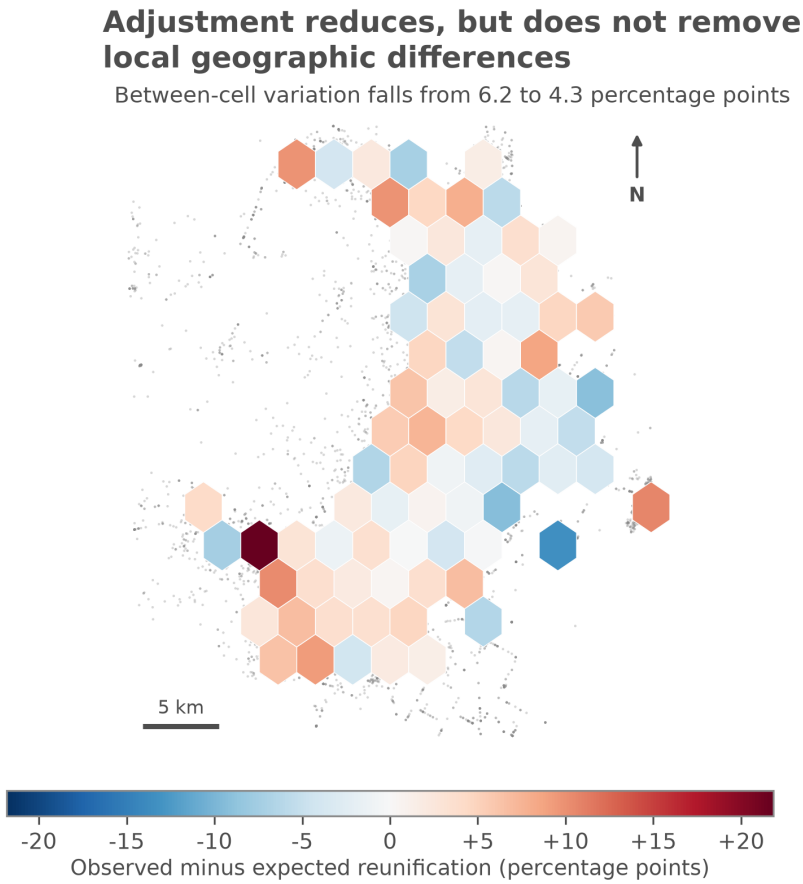


Figure 12: Gray points show geocoded pickups. Colored hexagons show observed-minus-expected reunification in cells with at least 60 dogs. Positive values mean higher-than-predicted reunification.

Check without ZIP boundaries. We also apply DBSCAN with haversine distance to 25,705 geocoded stray dogs after the same coordinate filtering used for the spatial map. Haversine is appropriate for latitude-longitude data because it measures geographic distance from those coordinates. We set `min_samples=60` to match the 60-dog minimum used for ZIPs and grid cells and choose ϵ = 450 m, approximately the median 60-neighbor distance. At 800 m, dense regions merge into one dominant urban-core cluster of about 20,900 dogs, so 450 m preserves more local structure. The 33 resulting hotspots cover 65.1% of dogs. Reunification ranges 10.5–40.8%, and altered share remains correlated with return rate (r = 0.76).

Impediments

1. **Geocoding is incomplete.** Intersections and landmarks are harder to geocode, so missing locations may not be random.
2. **Area patterns are not individual effects.** ZIP- and cell-level relationships are descriptive, and repeated addresses may reflect shared local factors.
3. **Ownership factors are missing.** Altered status may reflect veterinary access or prior ownership. Microchips and outreach are also unobserved. Without comparable Sonoma locations, the geographic result remains Austin-specific.

Future Work

- **Prospective test:** evaluate fixed classifier thresholds and different error costs on future shelter data.
- **More cities:** repeat the trait analysis across shelters to see how much the Pit Bull-type and black-coat associations vary by place.
- **Spay/neuter access:** combine tract-level Census measures with shelter data to test whether service access helps explain reunification.

Conclusion

The main lesson across all three problems is that raw patterns can change after adjustment or when tested on new data.

Prediction. Gradient boosting reaches Macro-F_1 0.588 versus a 0.163 majority baseline, but shelter-state features transfer poorly to Sonoma.

Adoption delay. Black coloring changes from a -2.9% raw difference to a $+10.2\%$ adjusted association, while the Pit Bull-type association is much larger in Austin than Sonoma.

Reunification. Altered share tracks ZIP-level reunification, but adjustment reduces between-cell variation by about 30%, and ZIP adds little held-out predictive value beyond measured dog traits.